

PHIL 350: Final

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1. Prove the following facts about entailments.
 - (a) $A \rightarrow B, \neg B \models \neg A$. (5 points)
 - (b) If $A \models C$ and $B \models C$ then $(A \vee B) \models C$. (10 points)
 - (c) If $A_1 \dots A_n \models C$ and $B_1 \dots B_k \models D$ then $A_1 \dots A_n, B_1 \dots B_k \models (C \wedge D)$. (10 points)
2. Determine whether the following arguments may be proven in the standard system (with axioms A1-A3). If can be, give the proof, if not show that there doesn't exist a proof by appealing to the soundness theorem.
 - (a) $\vdash (P \rightarrow (P \rightarrow Q)) \rightarrow (P \rightarrow Q)$. (10 points)
 - (b) $\vdash (P \rightarrow Q) \rightarrow Q$. (10 points)
 - (c) $\vdash (P \rightarrow Q) \rightarrow (P \rightarrow (R \rightarrow Q))$. (10 points)
 - (d) $\vdash (P \rightarrow (R \rightarrow Q)) \rightarrow (P \rightarrow Q)$. (10 points)
3.
 - (a) Define the connective $\Box$ (see truth table on last page of exam) using $\wedge, \vee$ and $\neg$. (5 points)
 - (b) Show that $\rightarrow$ and $\neg$ are together truth functionally complete. (10 points.)
 - (c) Show that $\uparrow$ (defined on last page) is truth functionally complete on its own. (7 points.)
 - (d) Show that $\wedge, \vee$ and $\rightarrow$ together are not truth functionally complete. (10 points).
4. This question is about a new proof system. It is formulated in the language $\mathcal{L}(\vee, \rightarrow, \neg)$ (so we have $\vee$ in addition to $\rightarrow$ and $\neg$). It has axioms:
 - B1. $A \rightarrow (A \vee B)$
 - B2. $(A \rightarrow B) \vee (B \rightarrow A)$
 - B3. $A \rightarrow \neg\neg A$

And two rules of proof

MP. You may infer B provided A and $A \rightarrow B$ have appeared earlier in the proof.

DI. You may infer $A \vee B$ provided A has appeared earlier in the proof.

We will use $\vdash'$ instead of $\vdash$ to indicate provability in this new system.

Prove that this system is sound: if there is a proof of B from $A_1 \dots A_n$ in this system then $A_1 \dots A_n \models B$. (15 points)

5. In this question we will show that the above system is not complete. Consider a non-standard notion of truth assignment $v_0 : Letters \rightarrow \{0, \frac{1}{2}, 1\}$. A sentence can have the value $\frac{1}{2}$ as well as 1 or 0. We will call this a *three-valued truth assignment*, and it may be extended to a *three-valued valuation* with the following recursive definition:

- $v(P) = v_0(P)$
- $v(A \vee B) = \max(v(A), v(B))$ (example: if $v(A) = \frac{1}{2}$ and $v(B) = 0$ then $v(A \vee B) = \frac{1}{2}$.)
- $v(\neg A) = 1 - v(A)$
- $v(A \rightarrow B) = 1$ if $v(A) \leq v(B)$ and $v(A \rightarrow B) = 0$ otherwise.
(PLEASE NOTE: this is different from the usual formula for $\rightarrow$!)

Review the following examples to check you've understood the definition. Examples: if $v(A) = 0$ and $v(B) = \frac{1}{2}$ then $v(\neg B) = 1 - \frac{1}{2} = \frac{1}{2}$. $v(A \vee B) = \max(0, \frac{1}{2}) = \frac{1}{2}$. And $v(A \rightarrow B) = 1$ since $0 \leq \frac{1}{2}$.

- (a) Show that the two rules of proof, MP and DI, preserve value 1 in a three-valued valuation: for instance, for DI you must show that if v is a three-valued valuation and $v(A) = 1$ then $v(A \vee B) = 1$. (5 points)
- (b) Prove the new axiom system is "sound" for the new notion of valuation. Show that if you can prove B from $A_1 \dots A_n$ in this system, and v is a three-valued valuation such that $v(A_1) = \dots = v(A_n) = 1$, then $v(B) = 1$. For this question, you may assume that any instance of the axioms, B1, B2 and B3, will have value 1 in any three-valued valuation (this is something I have checked for you). (10 points).
- (c) Use the soundness with respect to three-valued valuations to show that it is not the case that $\vdash' (P \vee \neg P)$ (there is no proof, in the new system, of $P \vee \neg P$ from no premises). (5 points)
- (d) Explain why this means that the system is not complete. (5 points).

P	Q	R	$\square(P, Q, R)$
T	T	T	T
T	T	F	F
T	F	T	F
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	T
F	F	F	F

P	Q	$P * Q$
T	T	F
T	F	T
F	T	T
F	F	T